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| **Fiscal Year:** | FY 2022 | **Task Last Updated:** | FY 10/19/2021 |
| **PI Name:** | Yokoyama, Hiroshi Ph.D. | | |
| **Project Title:** | Structure and Dynamics of Monodisperse Liquid Crystal Domains created on Suspended, Molecularly-Thin Smectic Films using Sub-Femtoliter Inkjet Technology | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | COMPLEX FLUIDS/SOFT MATTER--Complex Fluids | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **City:** | Kent | **State:** | OH |
| **Zip Code:** | 44242-0001 | **Congressional District:** | 13 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station |
| **Start Date:** | 12/30/2016 | **End Date:** | 12/29/2022 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA GRC |
| **Contact Monitor:** | Hatch, Tyler | **Contact Phone:** | 216.433.5073 |
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| **Flight Program:** | ISS | | |
| **Flight Assignment:** | Liquid Crystal Facility<br>NOTE: End date changed to 12/29/2022 per NSSC information (Ed., 1/26/22) | | |
| **Key Personnel Changes/Previous PI:** | October 2021 report: No change during this term. | | |
| **COI Name (Institution):** | Emelyanenko, Alexander Ph.D. ( Co-PI/ Lomonosov Moscow State University, Russia )<br>Tabe, Yuka Ph.D. ( Co-PI/ Waseda University, Japan ) | | |
| **Grant/Contract No.:** | NNX17AD68G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | 3D coarsening dynamics in emulsions, foams, and other non-equilibrium systems is an important and relatively well-studied problem. 2D coarsening dynamics of 1D interfaces is of increasing importance, but detailed studies are complicated by the lack of appropriate model systems. The proposed experiments on smectic films and bubbles offer a well-characterized, homogeneous platform for the study of 2D coarsening dynamics. The study of non-equilibrium molecular dynamics focused on the Lehmann rotation in an ideal condition is expected to provide definitive insight into the foundation of hydrodynamics of complex fluids and the dissipation mechanisms associated with molecular rotation, which is of central interest to develop new liquid crystals for industrial applications. The combination of theoretical simulations and the ground-based experiments as proposed here will elucidate the effect of microgravity on the self-organization of 2D fluid emulsions with embedded orientational degrees of freedom. Being molecularly thin, smectic films are highly susceptible to gravitational disturbances through meniscus forces and sedimentation of islands, as well as to thermal convection of the surrounding air. The knowledge to be attained by performing zero-gravity experiments in the International Space Station (ISS) will give us important clues to understanding the complex terrain of structure formation in complex soft matter. Two-dimensional fluids are experimentally attractive systems that are both accessible and complex enough to enable us to discover new routes to useful structures of emulsions in such fields as biological molecular recognition. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The principal scientific objective of this flight experiment is to explore the hidden phenomena in free standing smectic liquid crystal films in the microgravity environment in which the capillarity-induced forces disappear. Generally, interactions of microscopic particles and islands freely suspended in two-dimensional liquid films are of central significance in a wide range of industrial fields ranging from oil and mineral recovery, food processing, pharmaceuticals, coating and wet processes, as well as in basic sciences dealing with protein-protein interactions in cell membranes to name a few. In Earth's gravitational field, intricate molecular interactions are often overwhelmed by capillarity forces and are hardly accessible in direct physical experimentations. Microgravity in space enables us to approach these phenomena such as Ostwald ripening, molecularly mediated island-island interactions, and the Lehman rotation in liquid crystalline islands and films driven by the transmembrane molecular flow through the observation of configurational evolution of liquid crystalline islands on the smectic thin film. To prepare the required initial arrangement of islands, we develop and employ the sub-femtoliter inkjet deposition technology, which is integrated to an automated film preparation and observation hardware. Crucial for the liquid crystal science and technology is the understanding of intermolecular interactions responsible for the formation of liquid crystal phases. This space research aims to provide novel information that is hardly acquired from ground-based experiments. |
| **Task Progress:** | Liquid crystals are the unique state of matter in which the molecular order existing in the solid state can persist even in the flowing liquid state. Understanding the molecular interactions underlying the microscopic order is crucial for further development of better performing liquid crystals for industrial applications and also for elucidating the molecular mechanisms of a wide range of biological structures. The goal of this project is to study the evolution of microscopic islands of liquid crystals configured on a thin liquid crystal film in the microgravity environment. The islands are deposited on the film by means of a novel sub-femtoliter inkjet device in a prescribed configuration. The time dependent changes of configuration and the rotational motion of islands will reveal the hidden molecular action responsible for the liquid crystallinity.<br>The research effort in 2021, as in 2020, has also suffered seriously from the spread of COVID-19 infections across the nation since March 2020. In particular, laboratory activities were almost entirely stopped through the rest of the year. We have therefore focused our research efforts on perfecting the thickness measurement method that was started in 2019. As is well known, the thickness of free-standing films is quantized by the number of molecular layers and plays a significant role in determining the structure and interaction of droplets and their structural evolution. In the Lehmann rotation, the sense and speed of rotation undergo drastic variation as the thickness changes. Rapid and precise determination of the thickness distribution over the free-standing film is critical for the analysis of the experimental observations. We have developed a novel approach for thickness measurement based on the quantitative analysis of color of the reflected light, not on the intensity of reflection, which has been used for decades as the standard technique. For 2D analysis of the film images, the color is a much more reliable and robust variable than the intensity. By using a state-of-the-art high-resolution camera, it became possible to determine the thickness of the film with an accuracy better than 1 nm over the entire range of interest from less than 10 nm up to several hundred nanometers. This year, we have drastically improved the acquisition speed of the thickness measurement system by employing a lookup table scheme in place of the previous direct point-wise calculation. It is now possible to obtain the 2D thickness map of an area of the image consisting over one million pixels in one second. |
| **Bibliography Type:** | Description: (Last Updated: 02/05/2026) |
| **Articles in Peer-reviewed Journals** | Chen W, Yokoyama H. "Rapid thickness mapping of free-standing smectic films using colour information of reflected light II: real-time areal mapping using lookup table scheme." Liquid Crystals. 2021 Sep 7;49(3): 343-53. https://doi.org/10.1080/02678292.2021.1969458 , Sep-2021 |